

Name: Seth N, Richard B, Niko M, **Date:** 3/11/26 **Class:** 1

MyDesign Portfolio

Element C

Element C: Presentation and Justification of Solution Design Requirements

Doesn't Damage Mic #1 (HIGH Priority)

The clips need to secure the cable to the boom pole while minimizing stress on the mic/cable attachment piece, which will be achieved by making sure the mic angle does not change as a result of securing the cable.

Prior research revealed that a main problem with recording movies and short films is damage to the boom mic. This has been observed by Mr. Connolly and the film students, as many of the microphones have been damaged by tension from the cable.

Mr. Connolly agreed that the most important design requirement is to preserve the integrity of the boom mic, as it is an important and expensive piece for filming.

Weight #2 (HIGH Priority)

Design Requirement:

The boom pole cable clip must weigh no more than 3 grams so it does not affect the balance of the handling of the boom mic during filming.

Connection to Previous Research:

Research on microphone stand accessories shows that lightweight materials are important so equipment remains easy to handle while recording (Sweetwater). Heavy attachments can make boom poles harder to control, which could affect audio recording and filming quality.

Stakeholder Validation:

We shared this requirement with Mr. Connolly and several Film Festival students. They agreed that the clip must be very lightweight so it does not change how the boom pole

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feels when someone is holding it during filming. One student suggested keeping the clip “as light as possible,” which confirmed the need for a strict weight limit of around 3 grams.

Price #3 (HIGH Priority)

Design Requirement:

Each boom pole cable clip must cost no more than \$2 per unit so the school can afford to produce multiple clips for classroom use

Connection to Prior Research:

Research on classroom resource funding shows that schools often have limited budgets for equipment and supplies (National Education Association). Keeping the cost under \$2 ensures the design remains affordable for educational environments.

Stakeholder Validation:

Mr. Connolly confirmed that keeping the price low is important because multiple clips may be needed for different boom poles. After discussing the requirement, he agreed that a \$2 cost limit would make the clips practical for classroom use.

Design Requirement #4 (HIGH Priority)

Design Requirement:

The clip must hold the microphone cable within 1 centimeter of the boom pole so the cable stays out of the camera frame during filming.

Connection to Prior Research:

Research from the National Association of Broadcasters explains that proper cable management is important for maintaining professional production quality and preventing equipment interference during recording (National Association of Broadcasters)

Stakeholder Validation:

Film Festival students reviewed this requirement and agreed that keeping the cable very close to the boom pole would prevent it from appearing in camera shots. Mr. Connolly also confirmed that cable visibility is one of the main problems during filming.

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Design Requirement #5 (HIGH Priority)

Design Requirement:

The clip must attach securely to boom poles with a diameter between 0.75 inches and 1 inch, which are common sizes for boom microphones used in classrooms

Connection to Prior Research:

Industry guides for microphone stands and accessories show that boom poles typically fall within this diameter range, meaning the clip must fit standard equipment sizes (Sweetwater).

Stakeholder Validation:

Mr. Connolly confirmed that the boom poles used in the classroom are within this size range. He agreed that designing the clip to fit standard poles would make it usable with existing equipment.

Color #6 (LOW Priority)

Design Requirement:

The clip should be produced in a dark color such as black or dark gray so it blends in with the boom pole and is less visible during filming.

Connection to Prior Research:

Film production practices often use darker equipment to reduce reflections and visibility on camera, helping maintain professional-looking shots.

Stakeholder Validation:

Film students suggested using a darker color so the clip does not stand out on camera. Mr. Connolly agreed that darker colors would help keep the clip less noticeable during filming. However, it doesn't really matter because it will be off the screen.

Design Requirement #7 (LOW Priority)

Design Requirement:

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The clip should allow users to attach or remove it from the boom pole in less than 5 seconds without tools so students can quickly set up equipment during filming.

Connection to Prior Research:

Research about audio equipment maintenance shows that technicians must frequently adjust equipment quickly during recording sessions, making ease of use important (Bureau of Labor Statistics).

Stakeholder Validation:

Students who regularly set up filming equipment explained that quick setup is not that important as they primarily film outside of school hours, so they have plenty of time to set up to film and actually film.